

A Sentient Space Using Light Sensing with Particle Life

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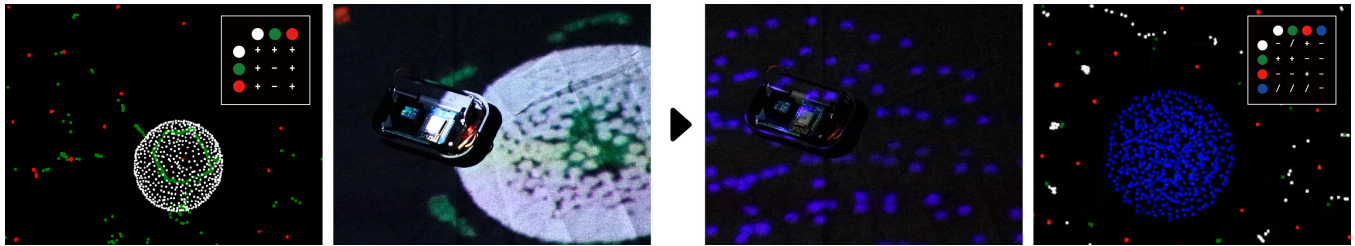


Figure 1: A cluster of white particles explores the projected environment. When it encounters the portable sensor, the system locates the device and generates blue particles that respond with adaptive behaviors.

Abstract

In interactive environments, the notion of space as a cognitive entity with the ability to autonomously perceive and explore its surroundings has yet to be fully developed. This paper presents a lightweight real-time interactive projection system that utilizes Particle Life algorithms and grid computing to enable spatial awareness and understanding of the internal environment, allowing for immediate projection adjustments in response to changes. The results show that interactive environments created by light as an information medium possess flexibility and broad application potential.

CCS Concepts

• Human-centered computing → User models.

Keywords

Light-Sensing, Cognition-Map, Particle Life Algorithm, Interactive Projection, Active Sensory Systems

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1 Introduction

If a space possesses active sensing capabilities, how will it understand its internal environment? Through sensing technologies, it can perceive the brightness of light, feel the flow of air and changes

in temperature. These external forces grant the space a form of "sensation," transforming it from a static structure into an organic entity that responds to environmental changes. Dana Cuff's "cyburg" concept [Cuff 2003] exemplifies such environments, where data from devices flow into the physical space, giving it perceptive and interactive capabilities, as demonstrated by MIT's ambientROOM prototype [Ishii and Ullmer 1998].

Focusing on light as a key environmental element, we recognize its potential as a vibrant medium for understanding, illustrated in projects like "Seventh Sense" [ultracompos 2011] and "Multiplex" [Rhizomatiks 2021], where light projections interact dynamically with performers. In this work, we propose a new method for achieving lightweight light perception and interaction within a projection environment. By utilizing a non-contact active sensing approach, we establish information transmission between the projection and the portable sensor, showcasing advantages in flexible data collection similar to echolocation.

2 Approach

We generate projection images using the Particle Life algorithm, which simulates the diverse social behaviors of organisms through simple color interaction rules, resulting in patterns like diffusion, movement, and convergence (see Figure 2). These behavioral expressions foster complex interaction dynamics. To connect Particle Life with the real world, we enhanced Hunar Ahmad's (2020) JavaScript

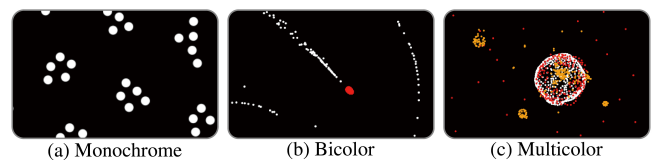


Figure 2: Monochrome clusters and bicolor to multicolor motion depict varying particle interactions.

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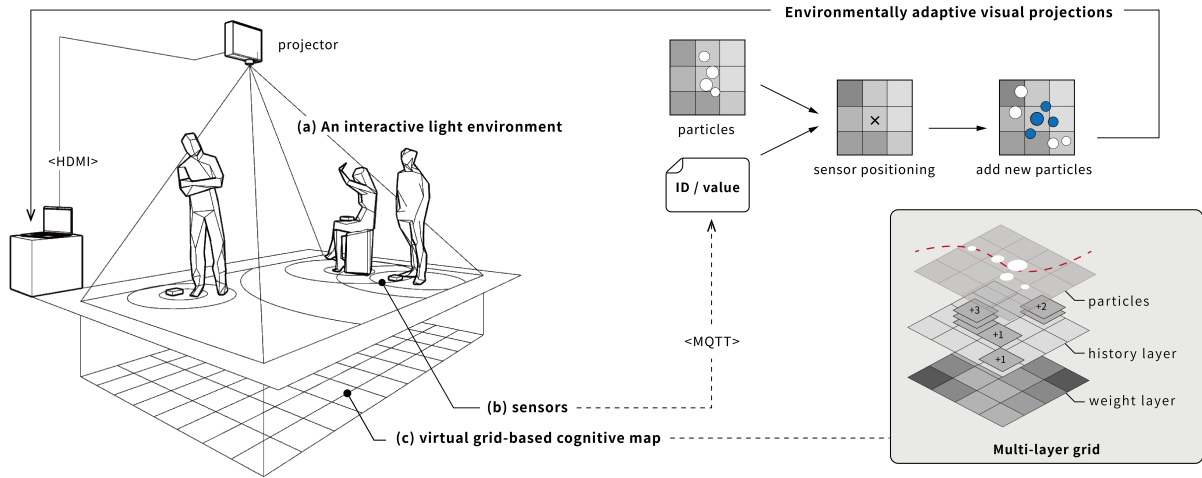


Figure 3: Workflow of the light-sentient space. (a) Generation of projection images using the Particle Life algorithm to create an interactive light environment. (b) Physical activities influence the sensing results of portable sensors, which wirelessly transmit data to the system via the MQTT protocol. (c) The system calculates the sensor's relative position within a multi-layered grid, triggering the dynamic generation or reduction of particles based on real-time environmental changes.

version of the algorithm by integrating MQTT communication and grid computing features.

First, the system adopts a multi-layer grid structure to track particle activity in real time. Each grid unit records the positions of specific particles (e.g., white particles), dynamically adjusting their weights to evolve the heatmap over time and visualize patterns of activity. Second, through heatmap recording, the system automatically compares data with information from external sensors, enabling positioning. As shown in Figure 3, when a sensor is illuminated by high-intensity light, the system locates the sensor at the grid coordinates where the white particles are most densely concentrated. The system records the sensor's ID and updates its coordinates accordingly to track dynamic movement. Lastly, once the sensor is positioned, the system triggers the generation of new particles at that location. The interaction forces between these newly generated particles, whether random or preset, lead to behavioral evolution. This process reflects how a space forms a cognitive map based on its internal activities and adapts its projection accordingly.

During testing, we categorized particle behaviors into exploration and interactive modes, applying different force rules. In

exploration mode, parameters favored white particles with wider motion ranges to efficiently locate portable sensors (see Figure 3). Once a response is triggered, the system switches to interactive mode, generating new blue particles at the location. These blue particles exhibit clustering, curiosity, or avoidance based on parameters, dynamically adjusting to the particle distribution and preventing an increase in the standard deviation (STD) caused by overly uniform behavior.

3 Conclusion

By extending the Particle Life algorithm and integrating MQTT communication with grid computing, this system successfully implements a real-time interactive projection system that quickly responds to physical activities and dynamically updates the projection content, offering flexibility and adaptability for various spatial interaction scenarios. Future work will focus on applying this prototype to multiple contexts and optimizing the cognitive system for enhanced interaction control.

Acknowledgments

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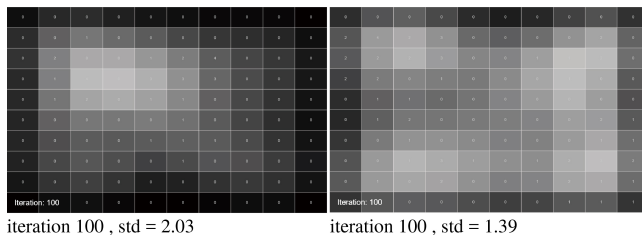


Figure 4: Parameter testing of exploration mode. Left: movement predominantly between diagonal areas. Right: a wider range of displacements.